

SIVA PRASATH K

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EDUCATION

SRM Institute of Science and Technology

Aug 2024 – May 2028

B.Tech, Electronics & Communication Engineering | CGPA: 9.485/10.0 (Top 5% of cohort)

Chennai, India

- Relevant Coursework:** Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, Discrete Mathematics, Microprocessors & Microcontrollers, Embedded Systems, Digital Electronics
- Problem Solving:** 300+ problems across LeetCode, Codeforces & GeeksforGeeks | LeetCode 1,500+ | Codeforces 1,200+ (Pupil). Graduating May 2028 — 4 semesters remain after the internship.

TECHNICAL SKILLS

Languages: C++, Java, Python, TypeScript, JavaScript, SQL, Bash

Backend & Systems: Node.js, NestJS, Express, FastAPI, PostgreSQL, MongoDB, Redis, BullMQ, REST APIs, WebSockets, Docker, Linux, CI/CD, JWT/Auth, Microservices, Event-Driven Architecture

Frontend: React, Next.js (App Router), Tailwind CSS, Zustand, React Query

Embedded & Hardware: ESP32, Arduino, Sensor Interfacing, I2C/SPI/UART, Real-Time Data Acquisition, Edge AI

Testing, Tools & ML: Jest, Vitest, Supertest, Playwright, Git, GitHub, Postman, PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLOv8

EXPERIENCE

Full Stack Developer Intern | [Certificate](#)

May 2026 – Jul 2026

Stallioni Net Solutions

Coimbatore, India

- Shipped a production government job portal for a Malaysian client on React, Node.js, Express and MongoDB — dynamic job listing, filtering and search, plus a multi-step application workflow with document upload and real-time status tracking.
- Designed the portal's secure admin panel to manage postings, review applications and export applicant data; implemented role-based access control (RBAC) over JWT authentication to meet client data-privacy requirements.
- Worked with senior engineers through code reviews and sprint planning, applying team standards for schema design, API versioning and automated testing.

Undergraduate Researcher — AI-Integrated IoT Healthcare

Aug 2025 – Present

SRMIST, supervised by Dr. Nithya V | Applied ML · Edge AI · Biomedical Systems

Chennai, India

- Built an ESP32-based IoT system for automated syringe injection with real-time sensor fusion and AI-driven control loops, hitting sub-5% dosage error using TensorFlow regression and time-series models.
- Prototyped NLP parsing of patient instructions via sequence-to-intent classification for voice-command caregiver interaction.

OPEN SOURCE CONTRIBUTIONS

Supabase (35k+ stars) | [TypeScript](#) | [PR #48053](#) · [Issue #48052](#)

- Audited packages/pg-meta's query-builder type layer and found a latent interface/implementation mismatch: `IQueryModifier.toSql` declared zero arguments while the implementing class and two live call sites already passed an options object controlling CTE vs. final-query SQL generation. Recognized it as the same bug class as a fix accepted upstream in a sibling file, and filed an issue with exact file/line evidence before opening the PR.
- Shipped a provably backward-compatible fix — widening the parameter to optional preserves every existing call — verified with a clean `tsc -noEmit` typecheck and a full unit-suite pass.

PROJECTS

AI Interview Preparation Platform | [NestJS](#), [Next.js 16](#), [PostgreSQL](#), [TypeORM](#), [Redis](#), [BullMQ](#), [Docker](#) | [Code](#)

- Solo-built end to end (NestJS REST API with DTO validation and JWT guards; Next.js 16 frontend on Zustand and React Query). Executes untrusted user code through a Docker-sandboxed runner with hard per-submission isolation — CPU, memory, process and network caps, read-only filesystem, non-root user and an enforced wall-clock timeout.
- Kept the API responsive under load by decoupling grading into a Redis-backed BullMQ queue (immediate 202 Accepted), bounding concurrent container execution independently of request traffic.
- Hardened auth against stolen-token replay using rotating server-side-hashed refresh tokens with reuse detection that revokes all sessions; validated via scripted replay, cross-account and CSRF/XSS attack simulations.

RoadGuard AI — Intelligent Road Safety Platform | [React](#), [TypeScript](#), [Node.js](#), [MongoDB](#), [Redis](#), [YOLOv8](#) | [Code](#)

- Built a geo-tagged hazard reporting platform; MongoDB 2dsphere indexing cut nearby-hazard lookups by ~90% and Redis caching reduced hot-endpoint latency by ~80%.
- Integrated a YOLOv8 microservice for road defect classification, with real-time hazard sync over WebSockets, Leaflet heatmaps and MongoDB aggregation dashboards.

ACHIEVEMENTS

- Innofusion Hackathon — Top 3 of 200+ teams:** built an AI therapist web app offering context-aware conversational support. | **Undergraduate Research Grant (2025)** for the AI-integrated IoT healthcare project at SRMIST.